

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech Branch : Mechanical/Mechanical(Sandwich) Semester : V

Subject Code & Name: BTMC502 Machine Design – I

Max Marks: 60

Date:03-01-24

Duration: 3 Hr.

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Use of Design data book is allowed.*
5. *Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Write short note on preferred sizes of component and make the table for all the series.		6
B) Explain basic Procedure of Machine Design		6
C) Which are the basic type of product forms.		6
Q.2 Solve Any Two of the following.		12
A) A mild steel shaft of 50 mm diameter is subjected to a bending moment of 2000 N-m and a torque T. If the yield point of the steel in tension is 200 MPa, find the maximum value of this torque without causing yielding of the shaft according to 1. The maximum principal stress		6
B) How to design the Knuckle joint		6
C) Design a cotter joint to support a load varying from 30 kN in compression to 30 kN in tension. The material used is carbon steel for which the following allowable stresses may be used. The load is applied statically. Tensile stress = compressive stress = 50 MPa ; shear stress = 35 MPa and crushing stress = 90 MPa.		6
Q. 3 Solve Any Two of the following.		12
A) State and explain stress concentration factor and its causes and remedies		6
B) Explain Goodman diagram.		6
C) Explain Reversed Stresses-Design for Finite and Infinite Life		6
Q.4 Solve Any Two of the following.		12
A) A solid circular shaft is subjected to a bending moment of 3000 N-m and a torque of 10 000 N-m. The shaft is made of 45 C 8 steel having ultimate tensile stress of 700 MPa and a ultimate shear stress of 500 MPa. Assuming a factor of safety as 6, determine the diameter of the shaft.		6
B) Explain spline keys		6
C) Design a cast iron protective type flange coupling to transmit 15 kW at 900 r.p.m. from an electric motor to a compressor. The service factor may be assumed as 1.35. The following permissible stresses may be used:		6

Shear stress for shaft, bolt and key material = 40 MPa

Crushing stress for bolt and key = 80 MPa

Shear stress for cast iron = 8 MPa

- Q. 5 Solve Any Two of the following.** **12**
- A) Explain Different forms of threads. **6**
- B) State the Advantages and disadvantages of welded joints. **6**
- C) A helical spring is made from a wire of 6 mm diameter and has outside diameter of 75 mm. If the permissible shear stress is 350 MPa and modulus of rigidity 84 kN/mm^2 , find the axial load which the spring can carry and the deflection per active turn. **6**

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